Chief Scientist Office 2026 Team Description Paper

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`https://www.robocup.ros-sier.com/`, `https://github.com/sbgisen`

Abstract This paper presents SoftBank Corp. Japan team Chief Scientist Office's mobile manipulator "SOAR" for RoboCup 2026 @Home Open Platform League.

1 Introduction

This paper details the RoboCup @Home Open Platform League team Chief Scientist Office from SoftBank Corp. Japan. This is our third appearance at RoboCup @Home Open Platform League having obtained 4th place in RoboCup 2024 @Home Open Platform League previously. Since our last participation in 2024, we have made significant improvements to our hardware and software system for mobile manipulator "SOAR".

At SoftBank Corp. Chief Scientist Office, we aim to pioneer a customizable and affordable commercial robot platform akin to the modular ease found in computer assembly or the historical T-Ford model. Our mission is to democratize robotics, making it accessible and adaptable for research and commercial use. We believe in a future where integrating robotics into business is as straightforward as selecting components for a custom PC build. With SOAR as a stepping stone, we aim to develop a versatile platform that celebrates modularity in both hardware and software, allowing seamless integration and upgrades through a suite of interchangeable modules.

1.1 Focus of research

Our research is motivated by the ambition to develop a customizable and affordable robot platform. We recognize the constraints of current technologies and are committed to exploring their limits and potential. Key areas of our interest include but are not limited to:

- Modularity and Configurability: Investigate how to design a system that allows for easy integration and customization of hardware modules.
- Device Integration: Tackling the hardware challenges related to the seamless integration of diverse devices and components

* The order of authors is based on age.

(a) SOAR. (b) Mobile Base. (c) Torso. (d) Manipulator. (e) Head.

Figure 1: SOAR and its four components.

We believe that these research areas are crucial to bridging the gap between the aspirational and the practical, laying the groundwork for a robot that can adapt to various needs.

2 Hardware Modules

SOAR is a mobile manipulator comprising four main components: the mobile base, the torso, the manipulator, and the head as shown in Fig. 1. The manipulator are commercially available products, while the mobile base, the torso, and the head are custom-developed by our team. The head and the torso are separate components designed for specific functionalities; we designed the head to accommodate various interchangeable sensors and the torso to enhance the manipulator's reach and versatility. All components are designed with minimal cable protrusion, with circuits centralized on the torso, thus streamlining the connection across components with minimal cabling. This modular approach not only facilitates customization but also organizes and simplifies maintenance and upgrades, allowing for the interchange of components as needed.

2.1 Mobile Base

SOAR's mobile base (Fig. 1b) is a compact 2-wheel differential drive platform, designed for indoor navigation. Both the driving wheels and casters are equipped with suspensions to ensure smooth movement and stability. The mobile base integrates a 2D liDAR and a 9-axis IMU for localization and autonomous navigation. It also includes battery management system and a docking charging connector, enabling the SOAR to recharge itself without human intervention thus extending its operational time. The motor drivers and the battery management system are connected to a USB hub, resulting in only two cables extending out: the USB cable from the device hub and the power supply cable.

Previously, SOAR used a commercially available mobile base with custom modifications. For RoboCup @Home 2026, we swapped with a newly developed, in-house base that has smaller footprint and equipped with a built-in LED button for direct user interface.

2.2 Torso

The torso (Fig. 1c), developed by our team, is a one degree of freedom linear actuator designed to vertically adjust the position of the head and manipulator. It incorporates a constant-force spring to provide gravity compensation, reducing motor load and improving energy efficiency during vertical movements. This mechanism provides an addition 350 mm of vertical clearance, significantly enhancing the manipulator's reach and versatility. The torso also serves as the central hub for SOAR's hardware components, housing the main PC and controller box.

2.3 Manipulator

The manipulator (Fig. 1d) is xArm 7[1], a commercially available product developed by uFactory. The end-effector is equipped with a two-finger gripper and a RGB-D hand camera.

2.4 Head

The head (Fig. 1e), designed as SOAR's primary sensory and interactive module, serves as a centralized platform for perception and user engagement. The module supports various range of sensors and user interface devices for environmental understanding and social interaction.

Equipped with a USB hub for efficient cable management and a separate USB for the high-bandwith depth camera, it allows for easy swapping of sensors and devices, ensuring that the component can be customized for specific requirements. The head extends three specific cables: a USB cable from the sensor hub, a separate USB for the depth camera, and a power supply cable.

In addition, the head has 2 degrees of freedom for pan and tilt movements, allowing SOAR to dynamically adjust its field of view for environment scanning, object tracking, and social interaction.

[1] `https://www.ufactory.cc/xarm-collaborative-robot/`

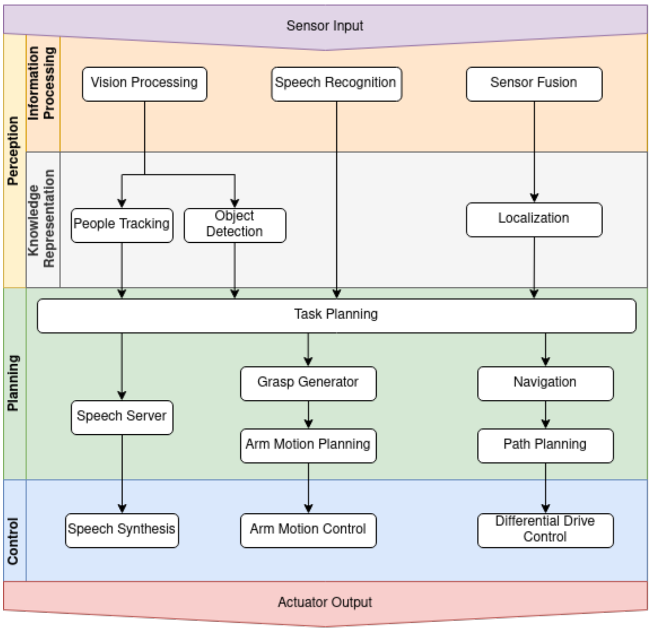


Figure 2: SOAR software overview.

3 Software

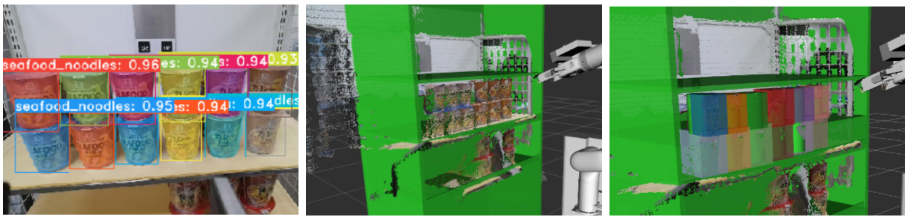
The overview of our software system, shown in Fig. 2 is comprised of three main layers:

1. The perception layer, a collection of software modules that take the robot's sensor data and process them to create a symbolic representation of the environment,
2. the planning layer, a collection of software modules that take the state of the environment to evaluate and generate a sequence of appropriate robot behaviors, and
3. the control layer, the collection of software modules that translates the planned behavior and sends appropriate commands to the actuators for the robot to interact with the environment

Previously, we used ROS1 Noetic as our primary software framework, but this year we migrated our software system to ROS2 Jazzy as ROS1 has reached its end of life.

3.1 Object Recognition

Object recognition is a crucial aspect of SOAR's perception system, enabling it to identify and interact with various objects in its environment. To achieve



(a) Detecting cup noodles with YOLOv8.

(b) Point cloud data and MoveIt planning scene.

(c) 3D collision boxes generated by jsk_pcl_ros package.

Figure 3: Object detection pipeline. YOLOv8's bounding box (Fig. 3a) and the point cloud data (Fig. 3b) are given to jsk_pcl_ros package to generate 3D collision boxes (Fig. 3c), which are passed to MoveIt planning scene.

robust and reliable perception, we combine 2D image-based object detection with 3D point cloud processing. We employed ultranlytics's YOLOv8[1] for real-time object detection, producing labeled bounding boxes for all recognized objects within the robot's field of view. These bounding boxes are then used to segment and cluster point cloud data within the detected regions to compute 3D collision boxes and object poses, which are essential for downstream manipulation tasks.

The pipeline is illustrated in Fig. 3.

3.2 Speech Recognition And Interaction

Our previous system relied on EfficientWord-Net for hotword detection, Google Dialogflow for conversational understanding, and Google Text-To-Speech for vocal responses. Our current system uses ChatGPT [2] for conversational understanding, known for its advanced natural language processing capabilities. The speech pipeline begins with Silero VAD [3] to segment incoming audio and Whisper [4] to transcribe speech to text. The transcribed text is then processed by ChatGPT [2] to generate a response, which is converted back to audio using its integrated text-to-speech capabilities.

In addition, we are exploring the use of the ChatGPT Realtime API, which supports low-latency, streaming interactions, and dynamic interaction where responses can be interrupted, updated, or redirected mid-generation based on new user input. To support this real-time conversational loop, we used Google's WebRTC AE3 echo canceller [**?**] to suppress audio feedback from the robot's own speakers, preventing them from being captured by the microphone. This enables continuous, overlapping dialogue where the user can speak while the robot is talking, resulting in more natural and fluid human–robot interaction.

3.3 Behavior

SOAR's behavior system is built on the SMACH ROS package, which provides a hierarchical state machine framework forming the core of the robot's decision-

Figure 4: Behavior for GPSR task.

Figure 5: Example of scene fitting a table.

making architecture. Each SMACH state is designed as a modular, reusable component, allowing behaviors to be composed, extended, and reorganized for different tasks. In standard operation, state transitions follow predefined logic, but for the GPSR (General Purpose Service Robot) RoboCup @Home task, we employ ChatGPT as a high-level planner. Given a natural-language command, ChatGPT generates a sequence of states to execute, which are then executed in order without relying on the traditional state transition conditions. This approach enables flexible task planning and adaptation to open-ended commands. For more controlled execution, certain states themselves encapsulate internal state machines, allowing for structured behaviors within a larger ChatGPT-defined plan. As a result, some states function as primitive actions, while others manage complex sub-behaviors, with ChatGPT determining their order of execution while the internal logic remains under our control.

The behavior pipeline along with its speech interaction for the GPSR task is illustrated in Fig. 4.

3.4 Scene fitting

In household environments, the scene around the robot is rarely static. While certain structures such as walls or large furniture remain fixed, many surfaces, such as tables and shelves where manipulation tasks occur, often shift slightly due to human activities. These subtle changes can introduce significant risks during manipulation, as the mismatch between the robot's internal scene model and the actual environment can lead to collisions. To address this, we developed a scene fitting process that aligns its internal scene model with the current environment before performing manipulation tasks.

During the mapping phase, SOAR records the point cloud and the pose of key furniture surfaces involved in manipulation tasks, such as tables and shelves. At runtime, the robot captures the point cloud of the same surface and applies a multi-stage ICP alignment to register the current observation with the previously stored model. The resulting transformation is used to update the robot internal scene model pose, ensuring that the motion planning accounts for slight environmental changes.

Fig. 5 illustrates an example of scene fitting for a table.

4 Contribution

SoftBank Corp. has continued to sponsor various robotic competitions[2][3] and conferences[4][5] to advocate human-robot interaction research. Our sister company, SoftBank Robotics Corp. is a long-term global sponsor for RoboCup, which further exemplifies our commitment to advancing robotic technologies and education. Despite our low direct scientific contribution, SOAR, our current prototype, represents our venture into supporting the research company and industry by offering a platform for research and study. While it is still in development, SOAR is designed with the future in mind, a testament to our initiative in paving the way for affordable and customizable robots, which we believe will make significant contributions to both academic research and practical applications.

5 Conclusions and future work

In this paper, we presented our currently developing robot SOAR which will participate in the RoboCup@Home League. We detailed the hardware specification and introduced several software modules, such as object recognition and speech recognition, which we will continue to refine to tackle the tasks in RoboCup @Home for our future work. By enhancing SOAR's capabilities in real-world interaction, we aim to contribute not just a flexible and affordable platform but also a robust platform for research and commercial use.

References

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[3] https://tsukubachallenge.jp/2023/organization/sponsors
[4] https://ac.rsj-web.org/2023/
[5] https://roscon.ros.org/jp/2023/

SOAR Hardware Description

Specifications are as follows:

- Dimensions: (w) 500 mm (d) 530 mm (h) 1440 mm - 1790 mm
- Weight: 113 kg
- Max Speed: 1.2 m/s
- Battery: 25.6V 40Ah LiFePO4, 3 hours operation time
- Mobile Base: differential drive
  - Sensors:
    - Hinson LE-50621 LiDAR
    - LDROBOT LD06
    - Wheeltec N100 9-axis IMU
- Torso unit: 1 DOF linear joint
  - PC: Intel NUC (NUC11PHKi7C)
- Manipulator: 7 DOF, 3.5 kg payload
  - Sensors:
    - ORBBEC Femto Bolt
- Head unit: 2 DOF pan-tilt
  - Sensors:
    - Azure Kinect DK RGB-D camera
    - NEEWER NW-410 condenser microphone
    - GENIUSPY JM800S-30-360X zoom camera
    - Seek Thermal CompactPRO Fast Frame
  - Display: THANKO C-79D21B
  - Speaker: YAMAHA VXS1MLB

SOAR Software Description

SOAR use the following open-source software components:

- Platform: Ubuntu 24.04 ROS2 Jazzy
- Navigation: Nav2, MPPI controller
- Manipulation: MoveIt2, MoveIt Task Constructor
- Object Recognition: YOLOv8
- Behavior Architecture: SMACH
- Speech Recognition: Silero VAD, Whisper
- Speech Generation: GPT-4o-mini-tts
- LLM: GPT-4o